

Rajesh Phulwaria

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LinkedIn — GitHub — LeetCode — Portfolio

Professional Summary

Aspiring AI/ML Engineer and research-oriented developer with hands-on experience in Machine Learning, Deep Learning, and Neural Network implementation using PyTorch and TensorFlow. Strong interest in understanding Artificial Intelligence from first principles through mathematical intuition, scratch implementations, experimentation pipelines, and practical deep learning workflows on Linux/Ubuntu environments. Skilled in Python, C++, DSA, and AI experimentation with growing focus on research-driven intelligent systems.

Education

- **Bachelor of Computer Applications (BCA)** 2024 – 2027
Aishwarya College, Rajasthan
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming
- **Higher Secondary Education** 2024
Board of Secondary Education, Rajasthan (RBSE)

Technical Skills

- **Languages:** Python, C++, SQL, JavaScript
- **AI/ML:** Machine Learning, Deep Learning, Neural Networks, CNNs, Regression, Classification, Model Evaluation, Feature Engineering
- **Frameworks & Libraries:** PyTorch, TensorFlow, Pandas, NumPy, Matplotlib, scikit-learn
- **Research & Data Workflows:** Data cleaning & Preprocessing, Exploratory Data Analysis (EDA), Experimentation Pipelines, Optimization Techniques
- **Computer Science Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, DBMS
- **Tools & Platforms:** Git, GitHub, Linux/Ubuntu, VS Code

Publications

A Comparative Analysis of Gradient Flow and Vanishing Gradient Effects in Deep Neural Networks 2026

Zenodo — DOI: 10.5281/zenodo.19351203

- Analyzed gradient propagation behavior in deep neural networks and compared activation functions to study vanishing gradient effects and training stability.

Projects

- **Foundational AI Lab — PyTorch, Deep Learning, Machine Learning** GitHub
 - Built a structured AI engineering and research repository focused on understanding Artificial Intelligence from first principles through mathematical intuition and scratch implementations.
 - Implemented practical deep learning workflows covering neural networks, CNN architectures, optimization techniques, Machine Learning concepts, and model evaluation using PyTorch.
 - Developed experimentation pipelines involving data preprocessing, exploratory data analysis (EDA), training workflows, and performance analysis for AI model experimentation.

- Developed and managed the AI experimentation environment on Linux/Ubuntu, improving familiarity with research-oriented development workflows, package management, and Python-based AI tooling.
- Maintained structured GitHub repositories documenting AI experiments, implementation notes, and deep learning workflows for continuous learning and research exploration.

● **Content-Based Movie Recommendation System — Flask, Machine Learning** GitHub

- Built a content-based movie recommendation engine using the TMDB dataset for personalized movie suggestions.
- Applied TF-IDF vectorization and cosine similarity techniques to compute movie similarity scores efficiently.
- Preprocessed metadata including genres, keywords, cast, and movie overviews for feature extraction and recommendation optimization.
- Deployed the recommendation engine as a Flask web application with an interactive user interface.

● **Neural Network from Scratch in C++ — C++, Deep Learning** GitHub

- Built a neural network from scratch in C++, implementing forward propagation, backpropagation, activation functions, and gradient descent optimization.
- Designed and trained a custom 2-2-1 neural architecture using tanh and sigmoid activation functions to solve the non-linearly separable XOR problem.
- Demonstrated core deep learning fundamentals and the importance of hidden layers through mathematical implementation without external ML frameworks.

Experience & Certifications

● **Programming in Python Internship – Internsforge** Feb 2026

- Completed hands-on internship focused on core and advanced Python programming concepts.
- Built mini projects involving automation, data manipulation, and Object-Oriented Programming concepts.
- Strengthened practical software development and problem-solving skills through project-based learning.
- Certificate ID: INTERNSFORGE-2026-0111

● **Deloitte Australia – Data Analytics Job Simulation** 2025

- Performed exploratory data analysis and derived business insights from structured datasets.
- Applied data-driven decision-making approaches in simulated consulting scenarios.
- Presented analytical findings in a structured professional reporting format.

Achievements

- Published AI research paper on gradient flow analysis in deep neural networks on Zenodo.
- Actively building AI/ML experimentation repositories and research-oriented projects on GitHub.
- Consistently developing foundational AI engineering skills through Linux-based deep learning workflows and scratch implementations.